

Synthesis and Luminescent Characterization of $\text{CaSO}_4\text{:Tm,Li}$ via Slow Evaporation, Coprecipitation and Solid-State Diffusion Routes

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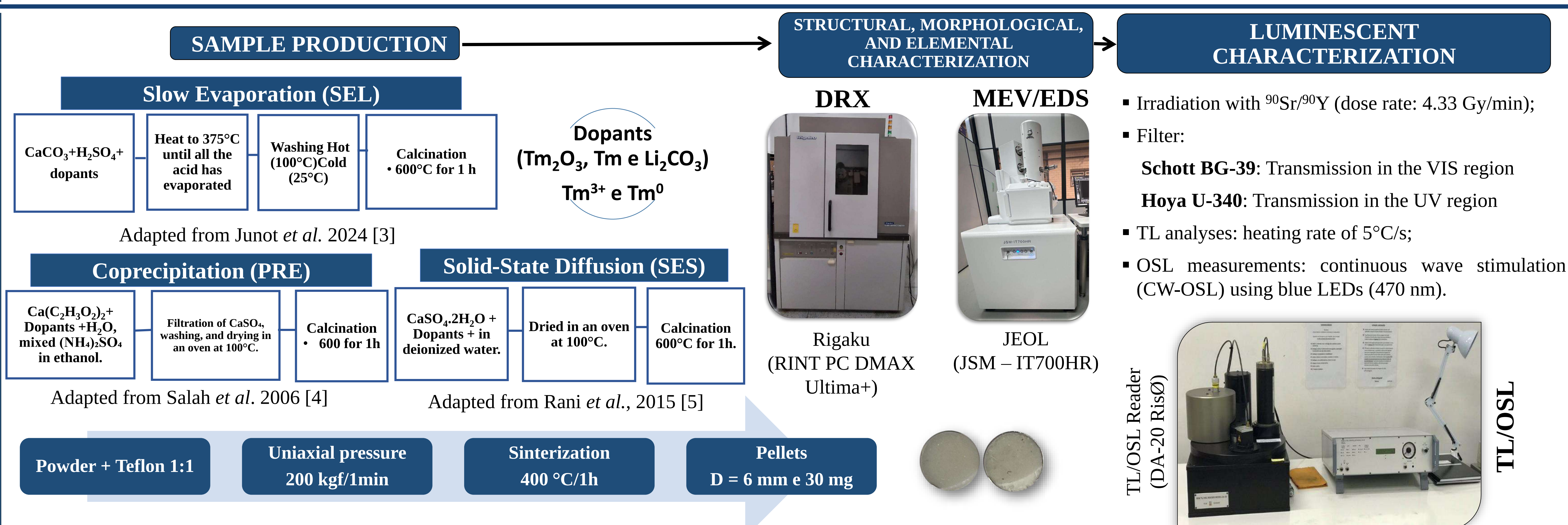
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Introduction

Calcium sulfate (CaSO_4) is widely used in various industrial sectors, particularly in ionizing radiation dosimetry, as exemplified by $\text{CaSO}_4\text{:Dy}$, a thermoluminescent dosimeter produced at IPEN and extensively utilized in Brazil [1,2]. This study investigated the thermoluminescent (TL) and optically stimulated luminescence (OSL) properties of CaSO_4 phosphors doped with thulium (in metallic and oxide forms) and codoped with lithium. The materials were synthesized via three distinct routes: slow evaporation, coprecipitation, and solid-state diffusion [3-5].

Materials and Methods



Results

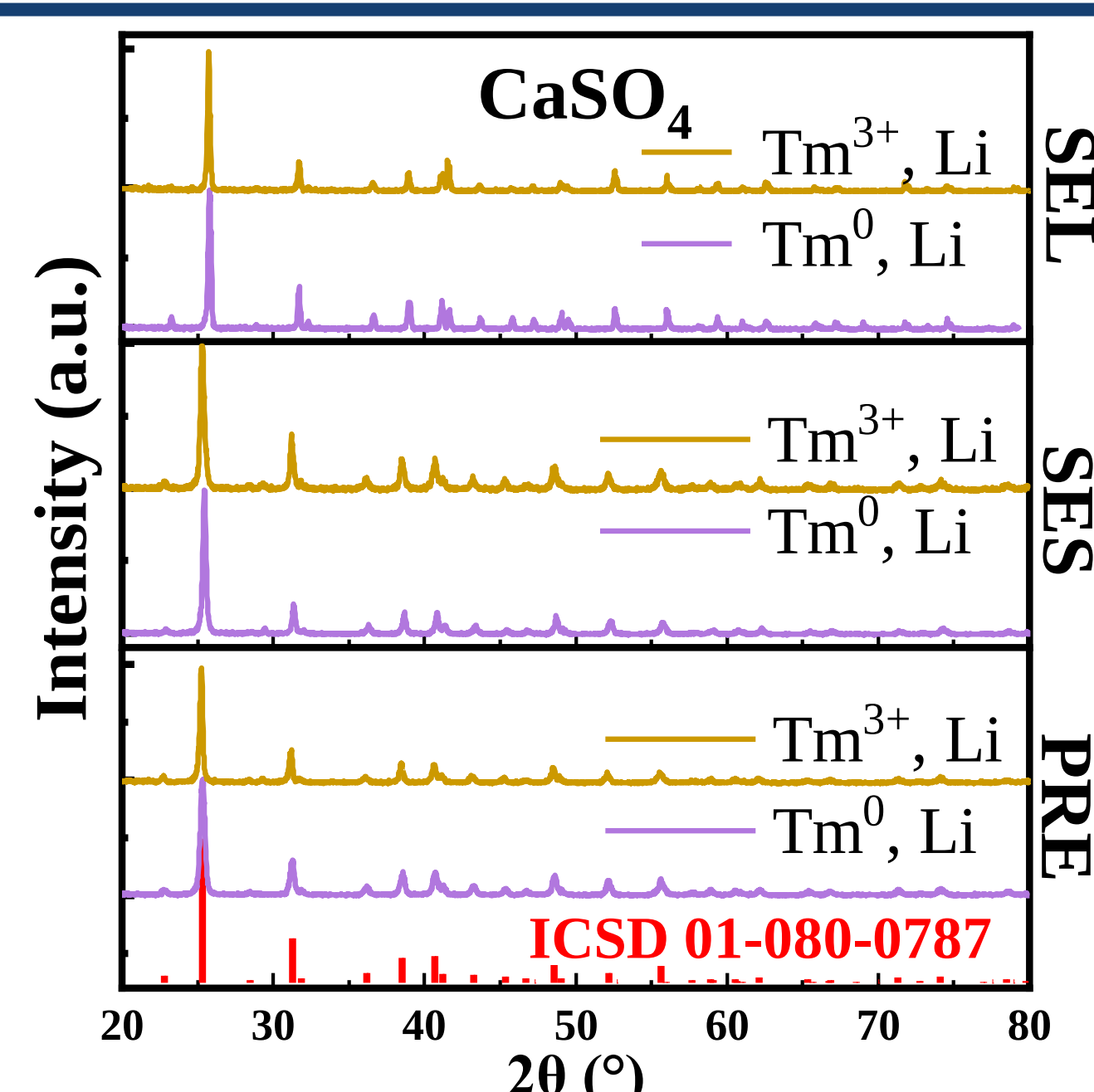


Fig. 1: X-ray diffraction patterns of $\text{CaSO}_4\text{:Tm}^{3+},\text{Li}$ powders produced by different synthesis routes.

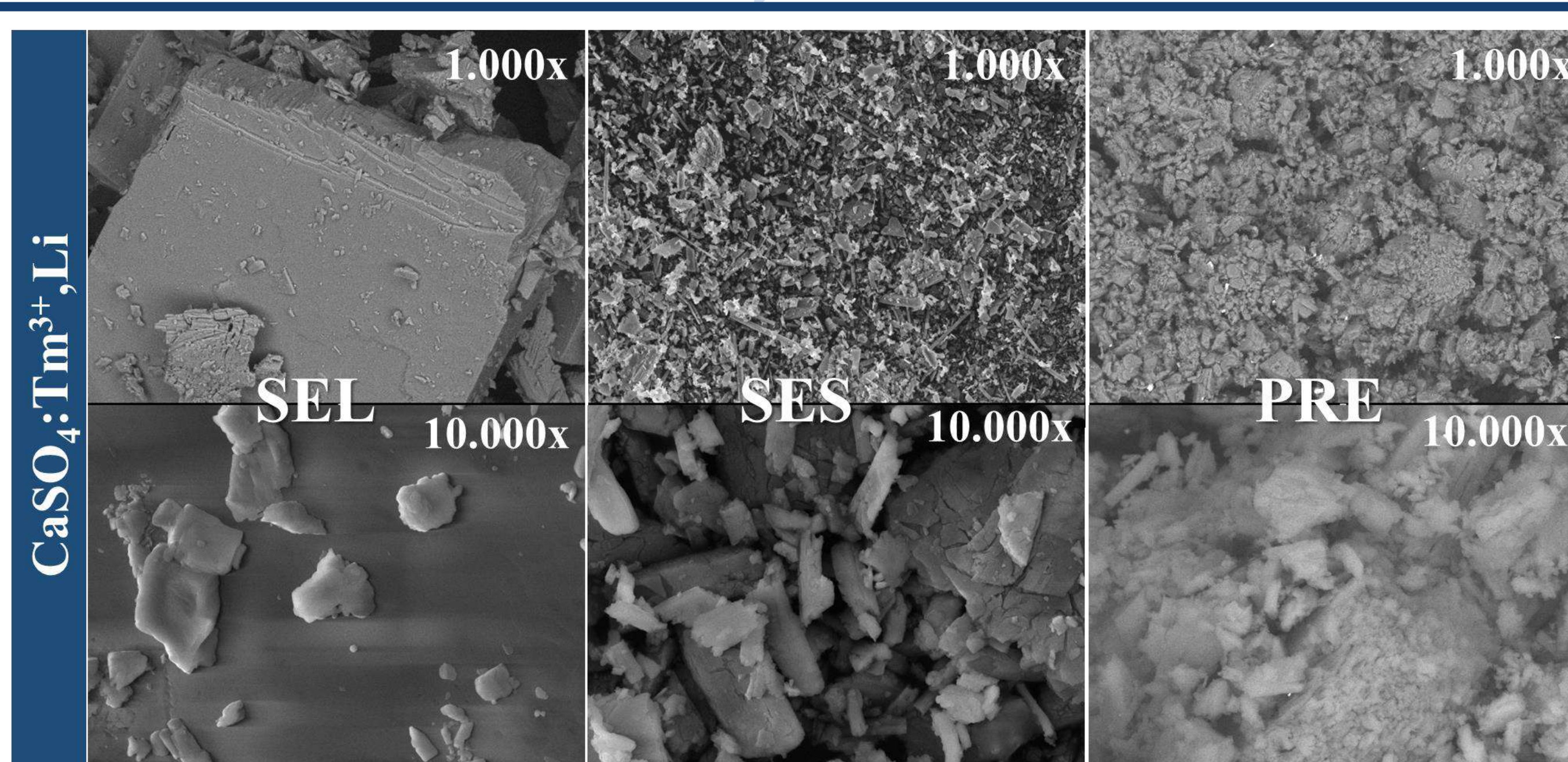


Fig. 2: SEM micrographs of $\text{CaSO}_4\text{:Tm}^{3+},\text{Li}$ polycrystals for the SEL, SES, and PRE routes.

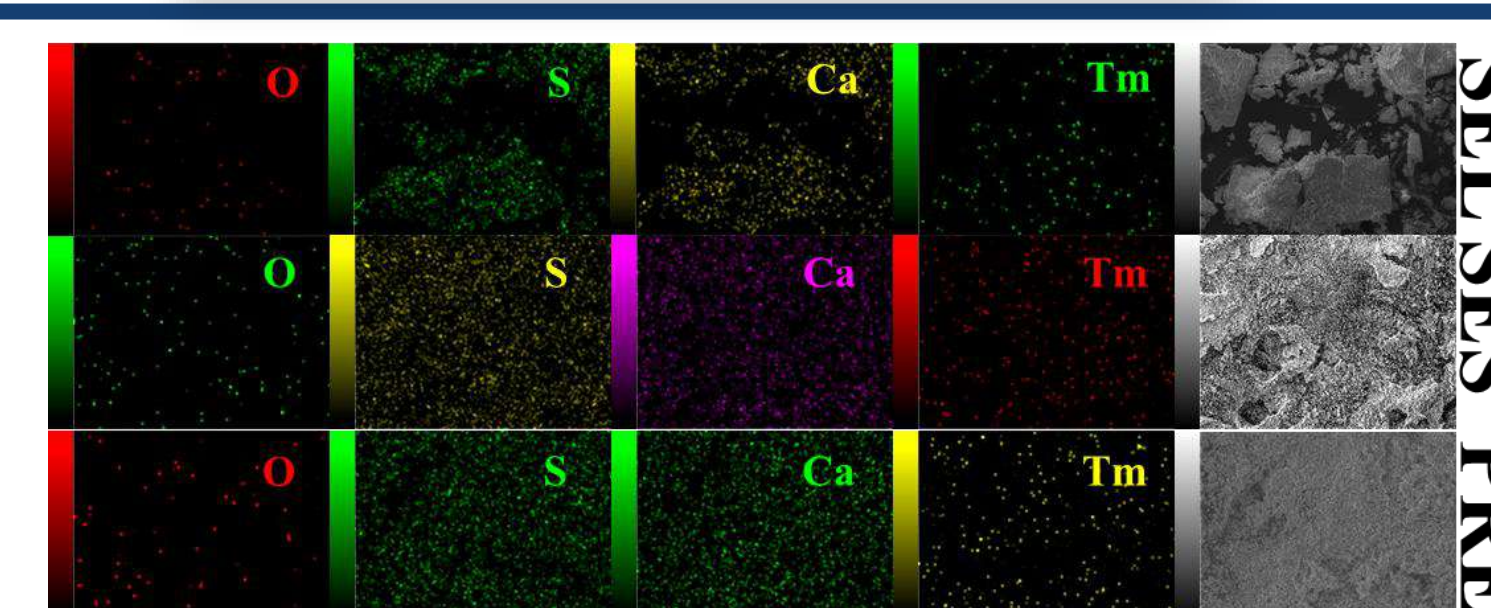


Fig. 3: EDS maps for $\text{CaSO}_4\text{:Tm}^{3+},\text{Li}$ by SEL, SES, and PRE routes.

Tab. 1: Atomic estimates obtained via EDS from the $\text{CaSO}_4\text{:Tm}^{3+},\text{Li}$ sample using the SEL, SES, and PRE routes.

Synthesis routes	O Atom(%)	S Atom(%)	Ca Atom(%)	Tm Atom(%)
SEL	44.62±0.84	9.73±0.17	10.24±0.24	0.04±0.01
SES	55.77±0.87	14.68±0.20	15.58±0.27	0.01±0.02
PRE	54.58±1.07	16.83±0.25	18.47±0.36	0.06±0.02

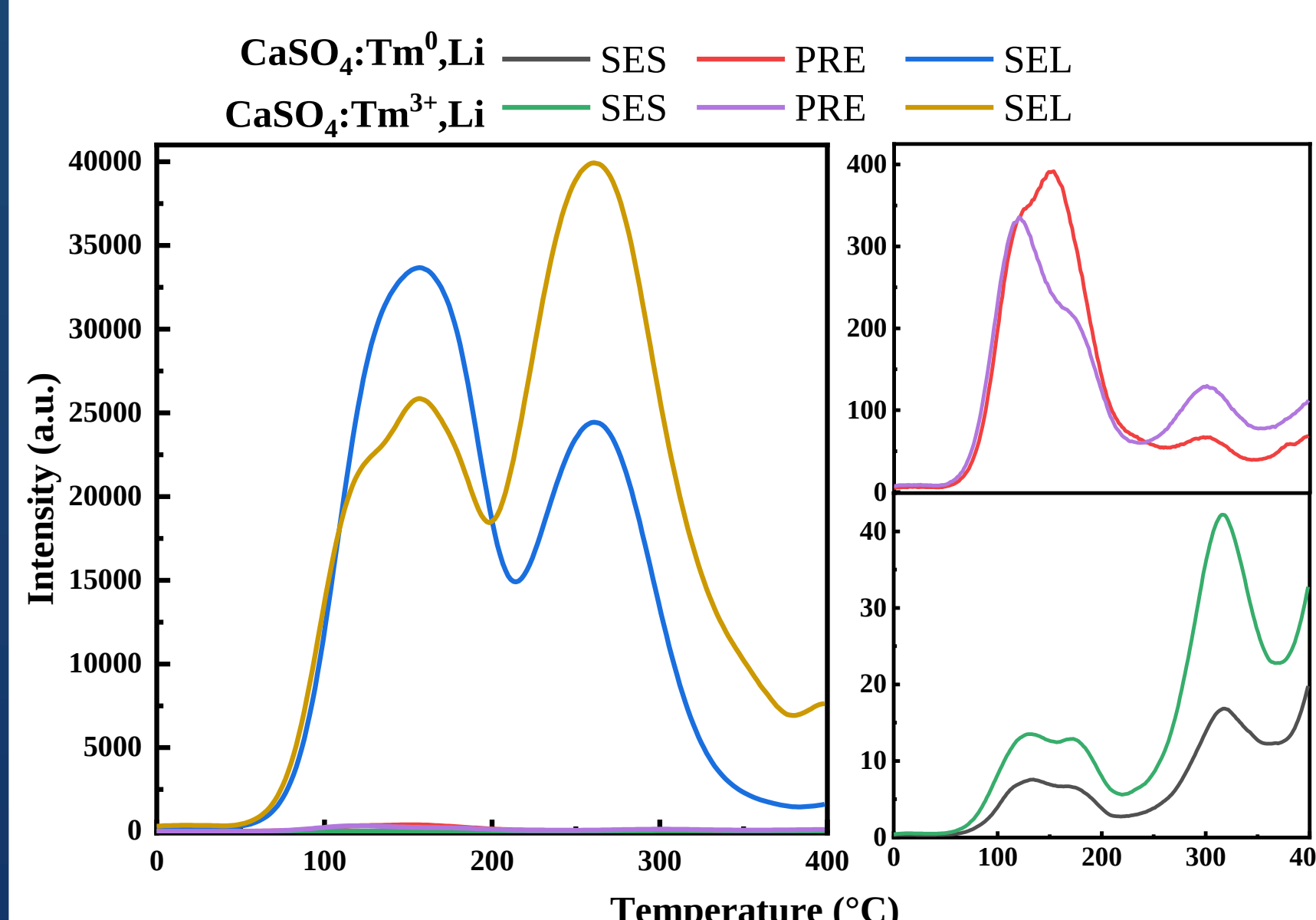


Fig. 4: TL curves of $\text{CaSO}_4\text{:Tm}^{3+},\text{Li}$ pellets produced by different routes, irradiated with 1 Gy of $^{90}\text{Sr}+^{90}\text{Y}$.

Tab. 2: TL signal obtained by integrating the area under the emission curve of $\text{CaSO}_4\text{:Tm}^{3+},\text{Li}$ samples produced by SEL, SES, and PRE.

TL	CaSO ₄ :Tm ³⁺ ,Li		CaSO ₄ :Tm ⁰ ,Li	
Route	Response (a.u.)	CV(%)	Response (a.u.)	CV(%)
SES	(3,6± 0,17) x 10 ⁵	4,8	(1,7±0,29) x10 ⁵	1,7
SEL	(4,3 ± 0,07) x 10 ⁸	1,6	(3,3±0,81) x10 ⁸	2,4
PRE	(3,0± 0,11) x 10 ⁶	3,8	(2,9±0,15) x10 ⁶	5,3

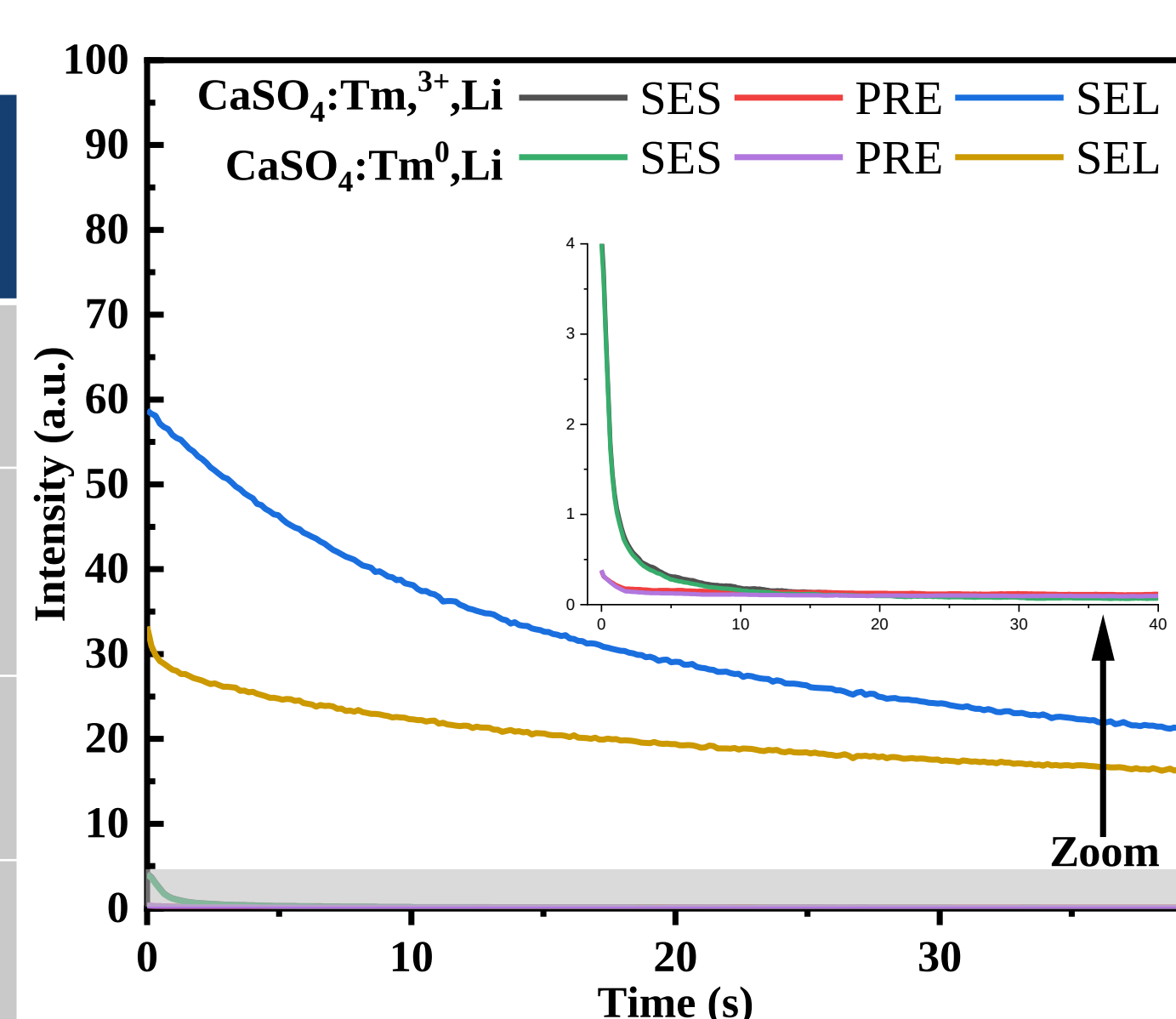


Fig. 7: OSL curves of $\text{CaSO}_4\text{:Tm}^{3+},\text{Li}$ pellets produced by different routes, irradiated with 1 Gy of $^{90}\text{Sr}+^{90}\text{Y}$.

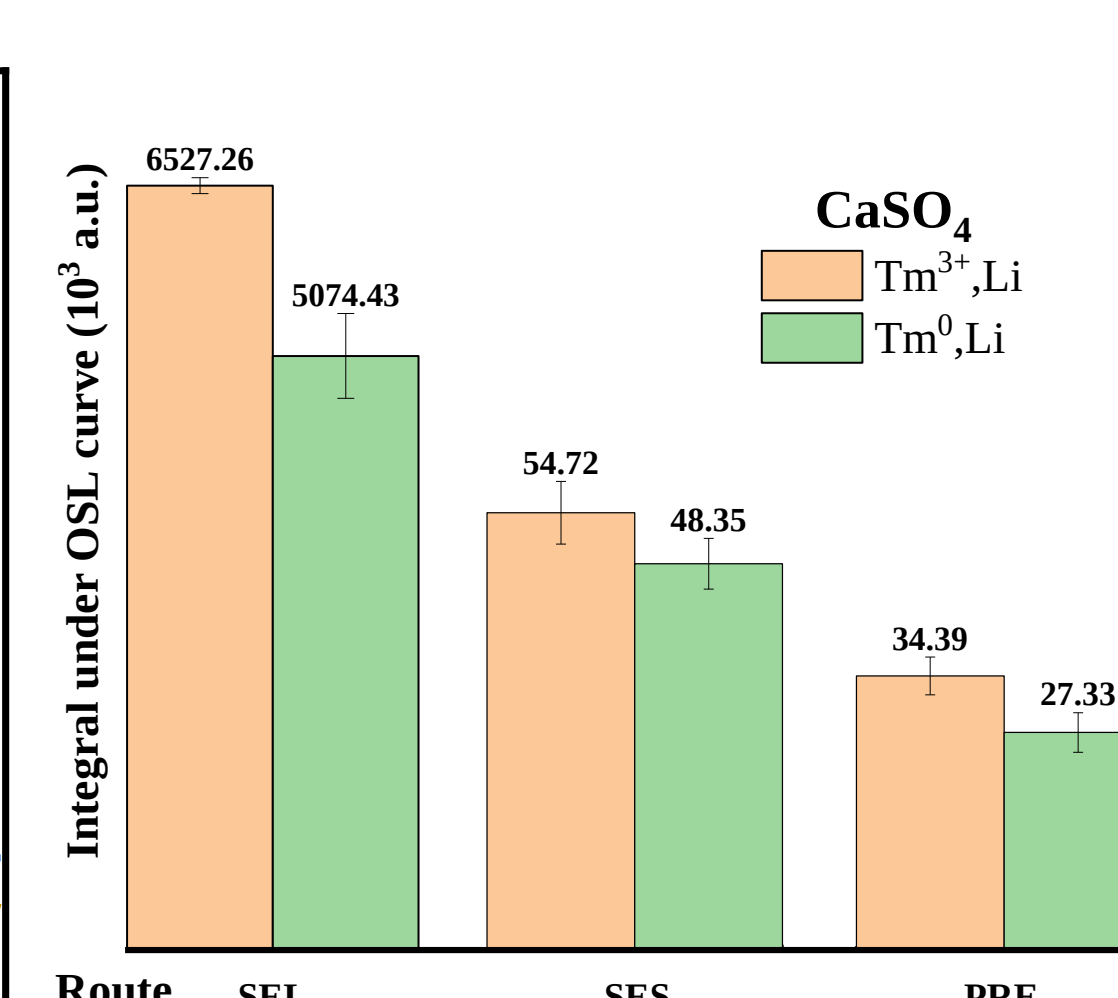


Fig. 8: OSL signal obtained by integrating the area under the emission curve of $\text{CaSO}_4\text{:Tm}^{3+},\text{Li}$ samples produced by SEL, SES, and PRE.

Conclusions

These preliminary results suggest that the $\text{CaSO}_4\text{:Tm,Li}$ phosphors synthesized in this work hold promising characteristics for future applications in ionizing radiation dosimetry, highlighting the need for complementary studies to comprehensively evaluate their dosimetric performance.

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